

UserManagerLib Unit Testing using NUnit

Introduction

This experiment explains how to write automated unit test cases for the UserManagerLib project using the NUnit framework. The CreateUser() method is tested to verify successful user creation as well as different exception scenarios related to the PANCardNo property.

Objectives

- Learn automated unit testing using NUnit.
- Understand how to test methods that throw exceptions.
- Verify successful user creation using valid input.
- Validate NullReferenceException and FormatException using Assert.That().

Implementation

Step 1: Create a Class Library project named UserManagerLib.Tests in the same solution.

Step 2: Rename the default test class to UserManagerTests.cs.

Step 3: Add the reference of UserManagerLib to the test project.

Step 4: Install NUnit and NUnit3TestAdapter using NuGet Package Manager.

Step 5: Create a test case with a valid 10-character PANCardNo and verify that the user is created successfully.

Step 6: Create a test case with a null or empty PANCardNo and verify that NullReferenceException is thrown.

Step 7: Create another test case with a PANCardNo that is not 10 characters long and verify that FormatException is thrown.

Step 8: Use Assert.That() and follow the Single Assertion Rule in every test case.

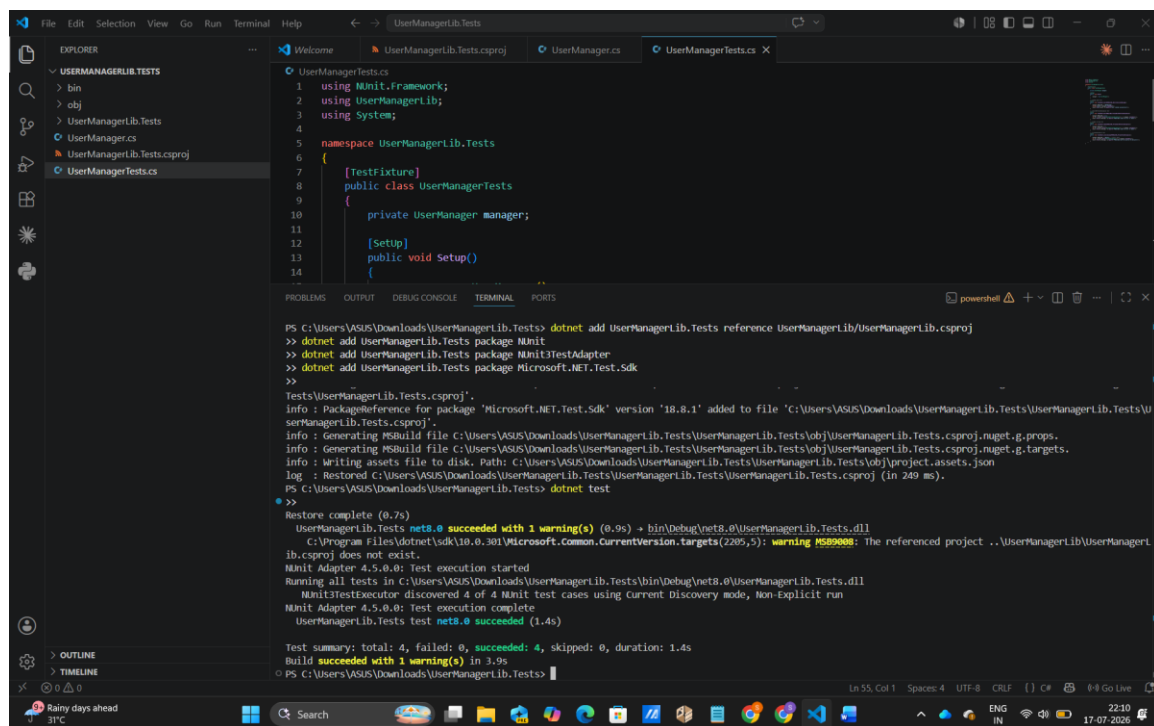
Step 9: Build the solution and execute all unit tests.

Step 10: Modify the CreateUser() method intentionally to break one of the test cases.

Step 11: Run the tests again and observe the failed result in Test Explorer.

Execution

Paste the screenshot of the successful NUnit Test Explorer or terminal output here.



```
PS C:\Users\VASUS\Downloads\UserManagerLib.Tests> dotnet add UserManagerLib.Tests reference UserManagerLib\UserManagerLib.csproj
>> dotnet add UserManagerLib.Tests package NUnit
>> dotnet add UserManagerLib.Tests package NUnit3TestAdapter
>> dotnet add UserManagerLib.Tests package Microsoft.NET.Test.Sdk
>>
Tests\UserManagerLib.Tests.csproj:
info : PackageReference for package 'Microsoft.NET.Test.Sdk' version '18.8.1' added to file 'C:\Users\VASUS\Downloads\UserManagerLib.Tests\UserManagerLib.Tests\UserManagerLib.Tests.csproj'.
info : Generating MSBuild file C:\Users\VASUS\Downloads\UserManagerLib.Tests\UserManagerLib.Tests\obj\UserManagerLib.Tests.csproj.nuget.g.props.
info : Writing assets file to disk. Path: C:\Users\VASUS\Downloads\UserManagerLib.Tests\UserManagerLib.Tests\obj\project.assets.json
log  : Restored C:\Users\VASUS\Downloads\UserManagerLib.Tests\UserManagerLib.Tests\UserManagerLib.Tests.csproj (in 249 ms).
PS C:\Users\VASUS\Downloads\UserManagerLib.Tests> dotnet test
>>
Restore complete (0.7s)
  UserManagerLib.Tests net8.0 succeeded with 1 warning(s) (0.9s) → bin\Debug\net8.0\UserManagerLib.Tests.dll
    C:\Program Files\dotnet\sdk\10.0.301\Microsoft.Common.CurrentVersion.targets(2205,5): warning MSB90008: The referenced project ..\UserManagerLib\UserManagerLib.csproj does not exist.
Running all tests in C:\Users\VASUS\Downloads\UserManagerLib.Tests\bin\Debug\net8.0\UserManagerLib.Tests.dll
  NUnit3TestExecutor discovered 4 of 4 NUnit test cases using Current Discovery mode, Non-Explicit run
  NUnit Adapter 4.5.0.0: Test execution complete
  UserManagerLib.Tests test net8.0 succeeded (1.4s)

Test summary: total: 4, failed: 0, succeeded: 4, skipped: 0, duration: 1.4s
Build succeeded with 1 warning(s) in 3.9s
PS C:\Users\VASUS\Downloads\UserManagerLib.Tests>
```

Expected Result

The application should create the user successfully when a valid 10-character PANCardNo is provided. It should throw `NullReferenceException` for null or empty input and `FormatException` when the PANCardNo length is invalid. All test cases should pass when the implementation is correct.

Conclusion

This experiment helped in understanding exception handling in unit testing using NUnit. Different test cases were created to verify both valid and invalid inputs, ensuring that the `CreateUser()` method behaves correctly under all conditions.